REPORT

Fast Food Nutritional Analysis and Calorie Prediction

GitHub Link: https://github.com/ayesha-haizoom/fastfood-data-science-project.git

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1. Introduction

Fast food is widely consumed because it is convenient and affordable. However, fast food often contains high levels of calories, fat, and sodium which may cause health problems such as obesity and heart disease.

This project aims to analyze the nutritional information of fast food items and identify patterns in calorie content.

Research Question

How are calories distributed among fast food items and how do they vary across different restaurants?

2. Data Acquisition

The dataset used in this project is the **Fast Food Calories Dataset** obtained from the TidyTuesday public dataset repository.

The dataset was loaded directly from GitHub using Python.

The dataset contains nutritional information of fast food items from several popular restaurants.

Key Variables

- Restaurant name
- Item name
- Calories
- Protein
- Sodium
- Total fat

These variables allow analysis of nutritional characteristics of fast food menu items.

	Unnamed: 0	restaurant	item	calories	cal_fat	total_fat	sat_fat	trans_fat	cholesterol	sodium	total
0	1	Mcdonalds	Artisan Grilled Chicken Sandwich	380	60	7	2.0	0.0	95	1110	
1	2	Mcdonalds	Single Bacon Smokehouse Burger	840	410	45	17.0	1.5	130	1580	
2	3	Mcdonalds	Double Bacon Smokehouse Burger	1130	600	67	27.0	3.0	220	1920	
3	4	Mcdonalds	Grilled Bacon Smokehouse Chicken Sandwich	750	280	31	10.0	0.5	155	1940	
4	5	Mcdonalds	Crispy Bacon Smokehouse Chicken Sandwich	920	410	45	12.0	0.5	120	1980	

Figure 1Dataset

3. Data Cleaning and Preprocessing

Steps:

Before analysis, the dataset was cleaned to improve data quality.

1. Remove unnecessary columns

Columns such as **Unnamed:0** and **salad** were removed because they were not useful for analysis.

2. Standardize text values

Restaurant names and item names were cleaned by removing extra spaces using string strip functions.

Example:

" Restaurant A " → "Restaurant A"

3. Handle missing values

Missing values were replaced with **0** to avoid calculation errors during analysis.

4. Remove duplicates

Duplicate rows were removed to ensure each record represents a unique menu item.

After cleaning, the dataset was ready for analysis.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand patterns in the dataset.

4.1 Calorie Distribution

This analysis shows the distribution of calories across fast food items.

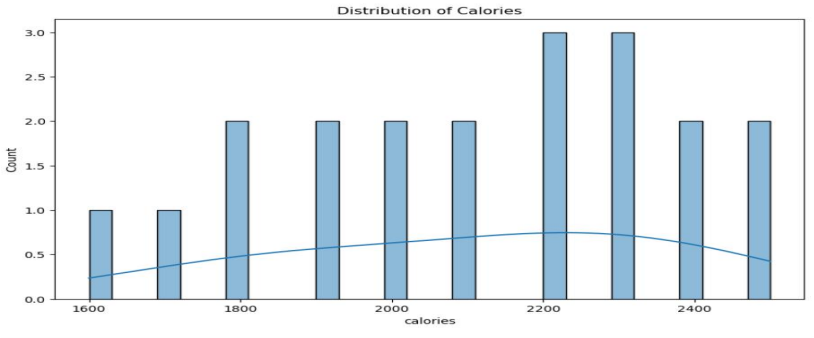

Figure 2 Distribution of Calories

➢ Explanation:

The histogram shows how calorie values are distributed among menu items. Most fast food items fall within a moderate calorie range, while some items contain extremely high calorie values.

4.2 Average Calories per Restaurant (Bar Chart)

This analysis compares the average calorie values across restaurants.

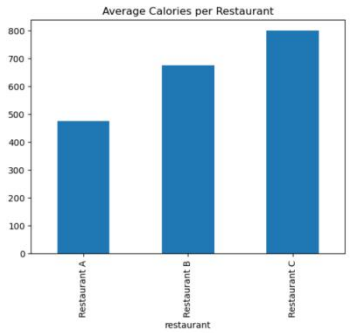

Figure 3 Average Calories per Restaurant

➢ Explanation:

The bar chart shows the average calorie content of menu items for each restaurant. This helps identify which restaurants tend to offer higher calorie foods.

4.3 Correlation Heatmap

A correlation heatmap was used to identify relationships between nutritional variables.

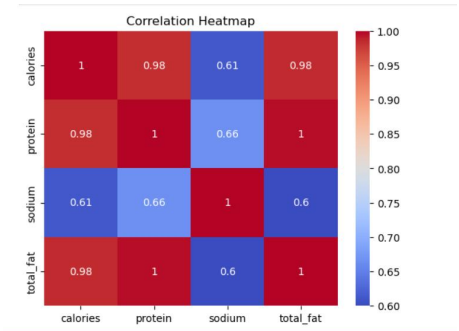

Figure 4 Correlation between Nutritional Variables

➢ Explanation:

The heatmap shows how different nutritional variables relate to each other. A strong positive correlation indicates that two variables increase together.

For example, calories often increase with higher fat or sodium levels.

5. ML / AI Models for Calorie Prediction

In this project, several machine learning and artificial intelligence techniques were used to analyze the relationship between nutritional values and calories in fast food items.

These methods help to predict calorie values and identify important nutritional factors affecting calories.

5.1 Linear Regression

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, r2_score

# Select features and target
X = df[['total_fat','protein','sodium','total_carb']]
y = df['calories']

# Train test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Train model
model = LinearRegression()
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Evaluation
print("MAE:", mean_absolute_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
```

Figure 5 Linear Regression Model

Linear Regression is a supervised machine learning algorithm used to predict continuous values. In this project, Linear Regression was used to predict **calorie values** using nutritional variables such as:

- Protein
- Sodium
- Total Carbohydrates

The dataset was divided into training and testing data using the train_test_split() function. The model was trained using the training data and predictions were made on the test data. The model performance was evaluated using **Mean Absolute Error (MAE)** and **R^2 Score**. A higher R^2 score indicates that the model can accurately predict calorie values.

5.2 Random Forest Regression

```
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

features = ['total_fat','protein','sodium','total_carb']

X_scaled = StandardScaler().fit_transform(df[features])

pca = PCA()
pca.fit(X_scaled)

plt.plot(pca.explained_variance_ratio_)
plt.xlabel("Principal Component")
plt.ylabel("Explained Variance")
plt.title("PCA Scree Plot")
plt.show()
```

Figure 6 Random Forest Regression

Explanation

Random Forest is an ensemble machine learning algorithm that uses multiple decision trees to improve prediction accuracy. In this project, Random Forest Regression was used to predict calories based on nutritional features. The algorithm builds many decision trees and combines their predictions to produce a more accurate result. Random Forest is useful because it can handle complex relationships between variables and reduce overfitting. The performance of the model was also evaluated using **R^2 Score and MAE**.

5.3 Principal Component Analysis (PCA)

```
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt

features = ['total_fat','protein','sodium','total_carb']

X_scaled = StandardScaler().fit_transform(df[features])

pca = PCA()
pca.fit(X_scaled)

plt.plot(pca.explained_variance_ratio_)
plt.xlabel("Principal Component")
plt.ylabel("Explained Variance")
plt.title("PCA Scree Plot")
plt.show()
```

Figure 7 5.3 Principal Component Analysis

Principal Component Analysis (PCA) is a dimensionality reduction technique used in machine learning.The dataset contains several nutritional variables, which makes analysis complex. PCA reduces the number of variables while preserving most of the information. In this project, PCA was applied after standardizing the data using StandardScaler(). The **Scree Plot** shows how much variance is explained by each principal component. The first few components explain most of the variance, meaning that fewer variables can represent most of the dataset information.

6. Reflection

Challenges Faced

- Cleaning the dataset by removing unnecessary columns and duplicates.
- Handling missing values in some nutritional variables.
- Understanding relationships between different variables such as calories, fat, protein, and sodium.
- Creating meaningful visualizations like histogram, bar chart, and heatmap.
- Implementing machine learning models such as Linear Regression and Random Forest in Python.
- Interpreting model evaluation metrics like MAE and R^2 score.

Key Learning Points

- Learned how to load and clean real-world datasets using Pandas.
- Understood how data visualization helps identify patterns and trends.
- Learned to use Matplotlib and Seaborn for creating graphs.
- Gained knowledge of machine learning models such as Linear Regression and Random Forest.
- Learned how PCA helps reduce data complexity.
- Understood how nutritional variables influence calorie values in fast food items.
- Improved skills in Python programming and data analysis.

7. References

McKinney, W. (2022). *Python for Data Analysis*. O'Reilly Media.

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